



ALLIANCE
UNIVERSITY
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NAAC
GRADE A+
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: BUSINESS SALES ANALYTICS DASHBOARD

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Experiment No.: 1

BUSINESS SALES ANALYTICS DASHBOARD

Problem Statement

Management needs a clear and interactive way to understand sales performance across products, categories, and regions. The raw business data contains valuable sales and profit information, but it is difficult to identify key trends and high-performing areas from the data alone. This project uses Power BI to transform the business dataset into an interactive dashboard that helps management analyze sales, profit, product performance, regional performance, and yearly trends for better business decision-making.

Methodology / Steps Followed

Data Collection – Collected the Business/Superstore dataset containing sales, profit, products, categories, regions, and order details.

Data Import – Imported the dataset into Microsoft Power BI.

Data Cleaning – Checked and prepared the data by verifying data types, missing values, and relevant fields using Power Query

Data Analysis – Created calculations for Total Sales, Total Profit, and Total Quantity.

Visualization – Created charts to analyze:

Sales by Region

Sales by Category

Sales Trend by Year

Top Products by Sales

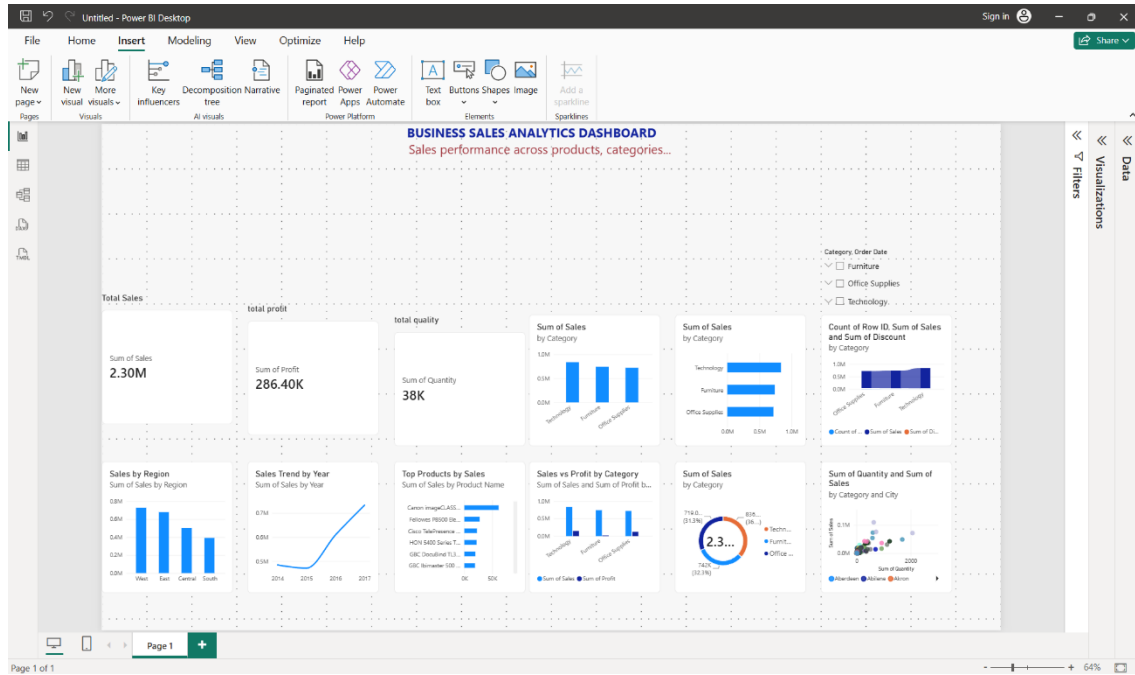
Sales vs Profit by Category

Interactive Dashboard – Added slicers for **Category and Order Date** to allow users to filter the analysis.

Business Insights – Compared sales and profit across products, categories, regions, and years to identify important business patterns.

Dashboard Presentation – Organized and formatted the visuals into an easy-to-understand Business Sales Analytics Dashboard for management decision-making.

Output



FileHomeHelpTable toolsMeasure tools

NameTotal Quantity

\$ - %

Data categoryUncategorized

Quick measure calculations

Home table

\$ - %

Auto

Quick measure calculations

Structure

Formatting

Properties

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Segment	Country	City	State	Region	Product ID	Category	Sub-Category	
65	CA-2015-135545	Tuesday, November 24, 2015	Monday, November 30, 2015	Standard Class	KM-16720	Consumer	United States	Los Angeles	California	West	OFF-PA-10003892	Office Supplies	Paper	Xerox 1943
161	CA-2016-162733	Wednesday, May 11, 2016	Thursday, May 12, 2016	First Class	TT-21070	Consumer	United States	Los Angeles	California	West	OFF-PA-10002751	Office Supplies	Paper	Xerox 1920
172	CA-2014-118962	Tuesday, August 5, 2014	Saturday, August 9, 2014	Standard Class	CS-12130	Consumer	United States	Los Angeles	California	West	OFF-PA-10000659	Office Supplies	Paper	Adams Phon
173	CA-2014-118962	Tuesday, August 5, 2014	Saturday, August 9, 2014	Standard Class	CS-12130	Consumer	United States	Los Angeles	California	West	OFF-PA-10001144	Office Supplies	Paper	Xerox 1913
559	CA-2016-113243	Friday, June 10, 2016	Wednesday, June 15, 2016	Standard Class	OT-18730	Consumer	United States	Los Angeles	California	West	OFF-PA-10003441	Office Supplies	Paper	Xerox 226
694	CA-2015-131534	Saturday, March 28, 2015	Thursday, April 2, 2015	Standard Class	AB-10165	Consumer	United States	Los Angeles	California	West	OFF-PA-10000743	Office Supplies	Paper	Xerox 1977
761	CA-2015-112319	Monday, April 21, 2015	Saturday, September 5, 2015	Standard Class	AB-10510	Consumer	United States	Los Angeles	California	West	OFF-PA-10003441	Office Supplies	Paper	Xerox 226
904	CA-2016-130477	Friday, April 8, 2016	Tuesday, April 12, 2016	Standard Class	LC-17140	Consumer	United States	Los Angeles	California	West	OFF-PA-10002947	Office Supplies	Paper	Xerox 1923
906	CA-2016-130477	Friday, April 8, 2016	Tuesday, April 12, 2016	Standard Class	LC-17140	Consumer	United States	Los Angeles	California	West	OFF-PA-10000919	Office Supplies	Paper	Xerox 1931
1144	CA-2014-146969	Monday, September 29, 2014	Friday, October 3, 2014	Standard Class	AP-10915	Consumer	United States	Los Angeles	California	West	OFF-PA-10002105	Office Supplies	Paper	Xerox 223
1569	US-2017-152002	Sunday, June 11, 2017	Same Day	Same Day	DV-13465	Consumer	United States	Los Angeles	California	West	OFF-PA-10003537	Office Supplies	Paper	White Dual P
1594	CA-2014-117337	Sunday, October 19, 2014	Sunday, October 19, 2014	Same Day	JF-15490	Consumer	United States	Los Angeles	California	West	OFF-PA-10004519	Office Supplies	Paper	Spiral Phone
1609	US-2016-115819	Tuesday, April 19, 2016	Sunday, April 24, 2016	Second Class	JO-15280	Consumer	United States	Los Angeles	California	West	OFF-PA-10003277	Office Supplies	Paper	Adams Telep
1683	US-2016-101497	Thursday, September 29, 2016	Saturday, October 1, 2016	Second Class	PS-18760	Consumer	United States	Los Angeles	California	West	OFF-PA-10000176	Office Supplies	Paper	Xerox 1887
1778	CA-2015-105347	Tuesday, November 24, 2015	Saturday, November 28, 2015	Standard Class	DP-13000	Consumer	United States	Los Angeles	California	West	OFF-PA-10000675	Office Supplies	Paper	Xerox 1919
2083	CA-2017-104647	Thursday, February 23, 2017	Wednesday, March 1, 2017	Standard Class	CK-12595	Consumer	United States	Los Angeles	California	West	OFF-PA-10002870	Office Supplies	Paper	Ampad Phon
2248	CA-2015-116092	Sunday, February 15, 2015	Wednesday, February 18, 2015	Second Class	JM-16195	Consumer	United States	Los Angeles	California	West	OFF-PA-10004285	Office Supplies	Paper	Xerox 1959
2252	CA-2015-116092	Sunday, February 15, 2015	Wednesday, February 18, 2015	Second Class	JM-16195	Consumer	United States	Los Angeles	California	West	OFF-PA-10000477	Office Supplies	Paper	Xerox 22
2413	CA-2014-151295	Wednesday, November 12, 2014	Sunday, November 16, 2014	Standard Class	JA-15970	Consumer	United States	Los Angeles	California	West	OFF-PA-10001947	Office Supplies	Paper	Xerox 1974
2492	CA-2014-113579	Saturday, December 13, 2014	Monday, December 15, 2014	Second Class	KD-16345	Consumer	United States	Los Angeles	California	West	OFF-PA-10001457	Office Supplies	Paper	White GlueT
2837	CA-2017-110842	Saturday, November 11, 2017	Thursday, November 16, 2017	Standard Class	GA-14725	Consumer	United States	Los Angeles	California	West	OFF-PA-10000520	Office Supplies	Paper	Xerox 201
2888	CA-2017-102099	Monday, December 18, 2017	Tuesday, December 19, 2017	First Class	EP-13915	Consumer	United States	Los Angeles	California	West	OFF-PA-10003172	Office Supplies	Paper	Xerox 1996
2889	CA-2017-102099	Monday, December 18, 2017	Tuesday, December 19, 2017	First Class	EP-13915	Consumer	United States	Los Angeles	California	West	OFF-PA-10000289	Office Supplies	Paper	Xerox 213
3422	CA-2016-134803	Saturday, May 28, 2016	Friday, June 3, 2016	Standard Class	CL-12565	Consumer	United States	Los Angeles	California	West	OFF-PA-10003790	Office Supplies	Paper	Xerox 1991
3557	CA-2017-118542	Friday, December 1, 2017	Tuesday, December 5, 2017	Standard Class	CC-12550	Consumer	United States	Los Angeles	California	West	OFF-PA-10004733	Office Supplies	Paper	Xerox 1905
3565	CA-2016-130029	Sunday, July 3, 2016	Wednesday, July 6, 2016	First Class	GT-14755	Consumer	United States	Los Angeles	California	West	OFF-PA-10000552	Office Supplies	Paper	Xerox 200
3945	CA-2016-133144	Monday, May 16, 2016	Saturday, May 21, 2016	Second Class	DO-13435	Consumer	United States	Los Angeles	California	West	OFF-PA-10004971	Office Supplies	Paper	Xerox 196
3971	CA-2014-135090	Saturday, November 22, 2014	Wednesday, November 26, 2014	Standard Class	SP-20920	Consumer	United States	Los Angeles	California	West	OFF-PA-10002245	Office Supplies	Paper	Xerox 1895
3985	CA-2016-135265	Thursday, July 7, 2016	Saturday, July 9, 2016	Second Class	CK-12370	Consumer	United States	Los Angeles	California	West	OFF-PA-10002923	Office Supplies	Paper	Xerox 1942
4118	CA-2015-153416	Tuesday, November 24, 2015	Sunday, November 29, 2015	Standard Class	TS-21340	Consumer	United States	Los Angeles	California	West	OFF-PA-10002005	Office Supplies	Paper	Xerox 225
4179	CA-2016-148593	Thursday, June 16, 2016	Saturday, June 18, 2016	Second Class	BD-11320	Consumer	United States	Los Angeles	California	West	OFF-PA-10001776	Office Supplies	Paper	Westbound h
4713	CA-2014-137020	Tuesday, December 9, 2014	Tuesday, December 13, 2014	Standard Class	MD-13000	Consumer	United States	Los Angeles	California	West	OFF-PA-10000111	Office Supplies	Paper	Xerox 1909

Table: total quantity (9,994 rows) Column: Total Quantity (6 distinct values)

Table: total quantity (9,994 rows) Column: Total Quantity (0 distinct values)

	Sales	Quantity	Discount	Profit
	146.73	3	0	68.9631
	5.98	1	0	2.691
	20.94	3	0	9.8418
	110.96	2	0	53.2608
	32.4	5	0	15.552
	33.4	5	0	16.032
	58.32	9	0	27.9936
	20.04	3	0	9.6192
	12.96	2	0	6.2208
	6.48	1	0	3.1104
17/8 x 11	122.97	3	0	60.2553
	13.44	3	0	6.5856
200/Messages	22.72	4	0	10.224
	18.97	1	0	9.1056
	368.91	9	0	180.7659
	37.44	6	0	16.848
	13.36	2	0	6.4128
	19.44	3	0	9.3312
	11.96	2	0	5.8604
	90.24	6	0	41.5104
	19.44	3	0	9.3312
	12.96	2	0	6.2208
	32.4	5	0	15.552
	182.72	8	0	84.0512
	45.36	7	0	21.7728
	12.96	2	0	6.2208
	17.34	3	0	8.4966
	53.82	9	0	24.219
	48.94	1	0	24.47
	19.44	3	0	9.3312
	46.35	5	0	21.7845
	10.44	2	0	5.3312

Data
Search
total quantity
Category
City
Country
Customer ID
Discount
Order Date
Order ID
Product ID
Product Name
Profit
Quantity
Region
Row ID
Sales
Segment
Ship Date
Ship Mode
State
Sub-Category
Total Profit
Total Quantity
Total Sales

Explanation of Data in the Dashboard

The dashboard is based on the Sample Superstore business dataset, which contains information about customer orders, products, sales, profit, quantities, regions, and order dates.

- **Total Sales:** Shows the overall sales generated from all orders.
- **Total Profit:** Represents the total profit earned from the sales.
- **Total Quantity:** Shows the total number of products sold.
- **Sales by Region:** Compares sales performance across different regions and helps identify the strongest-performing region.
- **Sales Trend by Year:** Shows how sales changed over different years and helps identify growth or decline.
- **Top Products by Sales:** Identifies the products that generated the highest sales.
- **Sales vs Profit by Category:** Compares sales and profit across product categories to identify profitable and high-performing categories.
- **Category Filter:** Allows management to view dashboard results for a specific category.
- **Order Date Filter:** Allows users to analyze sales for selected periods.

Overall, the dashboard converts raw business data into easy-to-understand visual information, helping management identify sales patterns, evaluate performance, and make better business decisions.

The final Power BI dashboard provides:

The final Power BI dashboard presents a clear and interactive overview of business sales performance. It provides key insights into total sales, total profit, and total quantity, while also analyzing sales across regions, categories, products, and years. Interactive filters allow users to explore the data based on category and order date. Overall, the dashboard helps management identify important business patterns, compare performance, and make data-driven decisions.

Conclusion

The retail sales data was successfully connected to Power BI Desktop from both CSV and MySQL data sources. The data was imported, explored, and analyzed using Power BI visualizations. A dashboard was created to analyze total sales, sales by store, product category, customer, and time. The final dashboard provides a clear and interactive view of sales performance and helps in understanding important retail business trends.

Learning Outcomes

After completing this project, the following learning outcomes were achieved:

- Learned how to import and manage business datasets in Power BI.
- Gained basic knowledge of data cleaning and transformation using Power Query.
- Learned to create measures and calculations using DAX.
- Developed skills in creating KPI cards, charts, and interactive slicers.
- Learned how to analyze sales, profit, products, categories, regions, and yearly trends.
- Developed the ability to identify key business patterns and performance indicators from data.
- Improved data visualization and dashboard design skills.
- Learned how Business Intelligence tools can support data-driven decision-making.

